

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No 9

Aim: Association Rule Mining

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their support and confidence.
- Interpret and analyze the discovered association rules.

Code:

```

frequent_itemsets.append(("frequent itemset (Pair)": pair, 'Support': round(support, 4))

freq_df = pd.DataFrame(frequent_itemsets).sort_values(by='Support', ascending=False)
print(f"--- Frequent Itemsets (Top 10, Support > 0.5) ---")
print(freq_df.head(10).to_string(index=False))
print("\n")

# --- REQUIREMENT: Generate Association Rules & Calculate Metrics ---
rules = []
for (item_a, item_b), count in pair_counts.items():
    support = count / max_trans
    if support > 0.005: # Lowered to 0.5% threshold
        conf_a_b = count / item_counts[item_a]
        conf_b_a = count / item_counts[item_b]
        rules.append(("Rule": f"{item_a} -> {item_b}", 'Support': round(support, 4), 'Confidence': round(conf_a_b, 4)))
        rules.append(("Rule": f"{item_b} -> {item_a}", 'Support': round(support, 4), 'Confidence': round(conf_b_a, 4)))

# Sort rules by highest confidence
rules_df = pd.DataFrame(rules).sort_values(by='Confidence', ascending=False)
print(f"--- Generated Association Rules (Top 10 by Confidence) ---")
print(rules_df.head(10).to_string(index=False))
print("\n")

# --- REQUIREMENT: Interpret and Analyze the Discovered Rules ---
print(f"--- INTERPRETATION & ANALYSIS ---")
print("Note: Since we adapted this for health data, rules indicate correlations between habits rather than purchased products.")
print("Highest Frequency (Support): The most frequent pairings will reveal the most common combinations of sleep and coffee habits.")
print("Highest Reliability (Confidence): A high confidence rule indicates that one specific habit strongly predicts another within the dataset.")
print("Cross-Category Selling: Here, this translates to how behaviors in one area (e.g., Physical Activity) strongly relate to outcomes in another (e.g., Sleep Quality).")
print("Core Item: Specific values (like 'Quality: Good') may act as anchor items, frequently appearing as the consequent in the highest-confidence rules.")

# Visualize Top 20 Rules
top_rules = rules_df.head(20)
plt.figure(figsize=(12, 10))
G = nx.DiGraph()

for _, row in top_rules.iterrows():
    a, b = row['Rule'].split(" -> ")
    G.add_edge(a, b, weight=row['Confidence'])

pos = nx.spring_layout(G, k=1.5, seed=42)
nx.draw(G, pos, with_labels=True, node_color='lightgreen', node_size=300,
        font_size=8, font_weight='bold', arrows=True)

# Add Confidence scores to the lines
edge_labels = {}
for k, v in G.edges():
    edge_labels[k] = (f"{v['Confidence']}: {v['Rule']}")
nx.draw_networkx_edge_labels(G, pos, edge_label=edge_labels, font_size=8)

plt.title("Top Apriori Association Rules by Confidence")
plt.show()

```

```

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Output:

